



GHIBRAN ESA ANUGRAH

Telecommunications Engineering | Politeknik Negeri Semarang

Semarang, Indonesia

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PROFESSIONAL SUMMARY

Telecommunications Engineering fresh graduate certified in Cisco CCNA, Fortinet NSE, and RF Engineering, with coursework in Cisco IoT & Digital Transformation. Hands-on experience at PT Telkom Akses in FTTx operations, enterprise SD-WAN deployment, and hardware troubleshooting. Proficient in IP routing/switching, embedded IoT (ESP32, C/C++), and web/database management (PHP, MySQL). Combines strong organizational leadership with field integration and rapid problem-solving skills.

EDUCATION

POLITEKNIK NEGERI SEMARANG

Diploma III in Telecommunications Engineering

GPA : 3.67 / 4.00

September 2023 – September 2026

SMK NEGERI 1 SLAWI

Computer and Network Engineering

July 2020 – April 2023

WORK EXPERIENCE

PT TELKOM AKSES

Intern Staff | September 2025 – December 2025

Tegal, Indonesia

- Executed emergency fusion splicing for a severed 144-core fiber optic cable at Jl. Gajah Mada Tegal, maintained OLT/ODC/ODP infrastructure, and performed OTDR testing to ensure 99%+ network availability.
- Deployed and configured Fortinet SD-WAN solutions for corporate clients (PT Indomarco Prismatama) to optimize enterprise network traffic and routing stability.

PT TELKOM AKSES

Intern Staff | June 2022 – September 2022

Slawi, Indonesia

- Provisioned, configured, and deployed ONT/STB devices with drop-cables for 30+ new IndiHome & IndiBiz subscribers while validating IPTV, VoIP, and broadband parameters using Optical Power Meter (OPM).
- Resolved onsite customer network incidents with a 95%+ SLA compliance rate by replacing damaged hardware and optimizing line parameters per operational standards.

PROJECT EXPERIENCE

Microstrip Antenna Design Projects | RF Engineering / CST Studio Suite | 2025

Designed and simulated single, 2x2, and 4x1 microstrip antenna configurations for the 902–928 MHz ISM band using CST Studio Suite, achieving optimal sub-GHz performance with S11 around -28 dB and VSWR of 1.1–1.3.

Radar Target Detection Simulator via Matched Filtering | Digital Signal Processing / GNU Octave | 2025

Developed a radar signal processing simulator integrating LFM chirp generation, AWGN interference, matched filtering, and peak detection algorithms, successfully estimating three simulated targets with zero range error.

IoT-Based Smart Camera Cabinet with Automated Hysteresis Climate Control | Internet of Things (IoT) & Embedded Systems / ESP32 | 2026

Designed and built an ESP32-powered camera storage cabinet with C/C++ firmware, MySQL database, automated hysteresis humidity control, UVC sterilization, and responsive web monitoring for mobile and desktop, achieving 99.21% regulation accuracy with stable operation exceeding 144 hours.

Water Resource Monitoring & Billing System | Web Application & Database / PHP, MySQL | 2025

Developed an integrated web application utilizing PHP, JavaScript, and MySQL for automated water resource monitoring, structuring real-time usage analytics, customer database management, and automated billing logic to optimize data processing.

ORGANIZATIONAL EXPERIENCE

KELUARGA MAHASISWA BIDIKMISI POLITEKNIK NEGERI SEMARANG

President | September 2024 – September 2025

- Directed organizational strategy and led an executive board of 50+ members to execute 13 academic programs, including securing strategic sponsorships with 8 corporate partners for 2 flagship events.

HIMPUNAN MAHASISWA ISLAM KOMISARIAT POLINES

General Secretary | June 2025 – Present

- Managed central administrative operations by drafting 105+ official letters, 2 major proposals, and event documentation, while coordinating 15+ executive meetings to enhance organizational reporting transparency.

CERTIFICATIONS

Fortinet Network Security Expert (NSE 1, 2, & 3)

Fortinet Training Institute (Scope: FortiGate Operator, Next-Gen Firewall, Cloud Security)

Radio Frequency Engineer - 193107501110-44/VSGA/BLSDM.Komdigi/2025

VSGA DTS 2025 BLSDM KOMDIGI Yogyakarta

Cisco Certified Network Associate (CCNA)

Cisco Networking Academy (Scope: Routing, Switching, Wireless Essentials & Introduction to Networks)

Introduction to IoT and Digital Transformation

Cisco Networking Academy

KKNI Level II – Computer and Network Engineering Competency - 61100 3513 2 0000258 2023

Badan Nasional Sertifikasi Profesi (BNSP)

AWARDS & ACHIEVEMENTS

- 1st Place – Sultan Agung Literacy Fest Essay Competition, Universitas Sultan Agung (2025)
- Best Paper – National Essay Competition, UNNES Scholarship Community (2025)
- Silver & Bronze Medals – Pekan Essay Nasional (PENA) 2, Universitas Dhyana Pura (2025)
- 2nd Runner-Up – Fair Essay Competition, FKMPI Politeknik Negeri Semarang (2025)
- 2nd Place – Information Network Cabling, Tegal Regency Student Competency Competition (2022)

SKILLS & COMPETENCIES

Telecom Infrastructure & RF	: Fiber Optic (FTTx, OLT, ODC, ODP), Optical Link Budget Analysis, Microstrip Antenna Design, RF Engineering, Cellular Networks (4G/5G NR), Wireless Networking
Embedded & Hardware Systems	: Microcontrollers (Arduino, ESP32), Sensors & Actuators, Circuit Design, System Integration, C/C++, MQTT, HTTP/API
Networking & Protocols	: TCP/IP, Subnetting, VLAN, Routing & Switching, OSPF, BGP, MPLS, VPN, Network Monitoring & Traffic Analysis, Network Troubleshooting
Software & Tools	: Cisco, MikroTik, Fortinet, Wireshark, Nmap, PRTG, Cacti, Proteus, GNU Octave, OptiSystem, CST Studio Suite, TEMS, Pathloss, Atoll, PHP, JavaScript, MySQL, Microsoft Office
Soft & Professional Skills	: Field Operation, Hardware Troubleshooting, Technical Documentation, System Integration & Testing, Project Coordination, Leadership

LANGUAGES : Indonesian – Native | English – Intermediate (POLINES TEPPS Score: 660)